

# $\phi$ -Dark Matter in SSEE-V3.6: Algebraic Mass Derivation and Two-Sector Proposal for the $f\sigma_8$ Growth Tension

SSEE Series — Paper 6

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## Abstract

The Structural Self-Energy Expansion (SSEE-V3.6) framework derives all cosmological parameters algebraically from the golden ratio  $\varphi$  and  $\pi$ , achieving a zero-free-parameter dark energy background (Papers 1–5) that has demonstrated strong coincidence with CMB, BAO, and cluster tests. Paper 5, using the CMB matter density  $\Omega_{\text{m,CMB}} = 0.309$  in the Poisson growth source, yields  $\sigma_8 = 0.834$ ,  $S_8 = 0.846$  ( $3.5\sigma$  KiDS-1000 and DES-Y3), and a mean  $f\sigma_8$  tension of  $0.70\sigma$  across six RSD surveys — essentially identical to  $\Lambda$ CDM in the growth-rate sector. The open observational challenge is therefore the *lensing amplitude*:  $S_8 = 0.846$  overshoots KiDS-1000 by  $3.5\sigma$ . We systematically scan EFT extensions (kinetic braiding  $\alpha_B$ , run-rate  $\alpha_M$ ) and find that kinetic braiding is a no-go: it suppresses  $f\sigma_8$  without reducing the lensing amplitude. We then propose a *two-sector* dark matter model in which  $\Omega_{\text{CDM}} = 0.160$  is supplemented by a  $\phi$ -dark matter component  $\Omega_{\phi\text{-DM}} = \Omega_{\text{m,CMB}} - \Omega_{\text{m,dyn}} = 0.149$ , active only on scales  $k < k_{\text{fs}}$ . The CMB matter density itself is a forward identity,  $\Omega_{\text{m,CMB}} = \omega_{\text{m}}/h^2$  with  $\omega_{\text{m}} = \omega_b + \omega_c + \omega_\nu$  and  $\omega_c = \text{KAL}_0 \omega_b n_s$  (Paper 1;  $0.41\sigma$  from Planck), so the two-sector split is a *difference* of two independent predictions, not a factor. The  $\phi$ -DM mass is a *forward prediction* written into a free scalar Lagrangian:  $m_\phi = \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 0.0685 \text{ eV} \times 594.28 = 40.70 \text{ eV}$ , where the multiplier is a *pure* ( $\varphi, \pi$ ) number and  $\Sigma m_\nu^{\text{active}} = \mathcal{R}_2 \times 0.9530 \text{ eV}$  with  $\mathcal{R}_2 = \Omega_{\text{DNAV}}/(\text{KAL} \cdot \text{TRIAL})$ . Nothing is fitted to data; the particle’s own temperature  $T_\phi = 0.539 T_\nu$  follows from the relic-abundance constraint. A direct two-sector CLASS Boltzmann run with this particle (zero fitted parameters) gives  $\sigma_8 = 0.747$  and  $S_8 = 0.758$  — in **0.04** $\sigma$  agreement with KiDS-1000 ( $0.759 \pm 0.024$ ), *resolving*

the  $S_8$  lensing tension from the  $3.5\sigma$  single-sector baseline. The free-streaming suppression scale is a CLASS *output*, not an input: a Viel fit to the particle/cold  $P(k)$  ratio returns  $\alpha = 1.117 \text{ Mpc}/h$ , corresponding to  $k_{\text{fs}} = 0.754 h/\text{Mpc}$  — the primary falsifiable prediction for DESI Y3 / Euclid (2026–2028). Because  $m_\phi \simeq 41 \text{ eV}$  makes the particle non-relativistic at  $z_{\text{nr}} \sim 10^5 \gg z_{\text{rec}}$ , it behaves as cold dark matter for the CMB: the acoustic peaks shift by only  $\sim 1 \ell$  relative to  $\Lambda\text{CDM}$ . The same free-streaming that resolves  $S_8$  also reaches the  $\sigma_8(R=8)$  window probed by RSD, so the two-sector  $f\sigma_8$  tension rises modestly to  $0.93\sigma$  (from the  $0.70\sigma$  single-sector baseline), still  $< 1\sigma$  and close to  $\Lambda\text{CDM}$  ( $0.73\sigma$ ): the deliberate cost of lowering  $\sigma_8$  to fix the lensing amplitude. We explicitly note that the reported TT residual is computed against our own CLASS  $\Lambda\text{CDM}$  rather than a full Planck `plik` likelihood with covariance; a paper-grade CMB validation is deferred.

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# 1 Introduction

The Structural Self-Energy Expansion (SSEE-V3.6) framework [Almeida Vallejo, 2026a] is a zero-free-parameter dark energy model in which the equation-of-state parameters  $(w_0, w_a) = (-0.840, -0.670)$  are derived algebraically from  $\varphi$  (golden ratio) and  $\pi$ . The framework has passed four independent observational tests:

1. **Paper 2** [Almeida Vallejo, 2026b]: DESI DR2 [Abdul-Karim et al., 2025] BAO ( $0.24\sigma$  Pantheon+) and galaxy cluster mass discrepancies ( $\chi_r^2 = 0.122$ ).
2. **Paper 3** [Almeida Vallejo, 2026c]: CMB Planck PR4 (plik\_lite  $\Delta\text{BIC} = -32.9$   $\omega_m$ -direct (full plik MCMC;  $-23.9$  plik\_lite), in favour of SSEE-V3.6 vs.  $\Lambda\text{CDM}$ ).
3. **Paper 4** [Almeida Vallejo, 2026d]: Algebraic derivation of  $n_s$ ,  $H_0$ ,  $\Omega_m$ ,  $\Omega_b h^2$  from the Nine Sovereignties.
4. **Paper 5** [Almeida Vallejo, 2026e]: Israel-Stewart [Israel and Stewart, 1979] causal gradient stability ( $c_{s,\text{eff}}^2 = 0$  exact), growth factor  $\mathcal{G} = 1.0032$  (slight enhancement vs.  $\Lambda\text{CDM}$ ),  $\sigma_8 = 0.834$ ,  $S_8 = 0.846$  ( $3.5\sigma$  KiDS; single-sector open lensing tension).

## 1.1 The residual tension

Paper 5 corrected the growth equation to use the CMB matter density  $\Omega_{\text{m,CMB}} = \omega_m/h^2 = 0.309$  in the Poisson source, yielding  $\mathcal{G} = D_1^{\text{SSEE}}/D_1^{\Lambda\text{CDM}} = 1.0032$  (slight enhancement),  $\sigma_8 = 0.834$ , and a mean  $f\sigma_8$  tension of  $0.70\sigma$  across six RSD surveys [Beutler et al., 2012, Howlett et al., 2015, Alam et al., 2017, Hou et al., 2021] — essentially the same as  $\Lambda\text{CDM}$  ( $0.73\sigma$ ). The growth-rate sector is therefore already consistent at the single-sector level; the two-sector free-streaming introduced below lowers  $\sigma_8$  to resolve  $S_8$  and, since that suppression reaches RSD scales, raises the  $f\sigma_8$  tension modestly to  $0.93\sigma$  (still  $< 1\sigma$ ).

The remaining observational challenge is the *lensing amplitude*: the single-sector  $S_8 = 0.834 \times \sqrt{0.309/0.3} = 0.846$  lies  $3.5\sigma$  above KiDS-1000 ( $0.759 \pm 0.024$ ) [Asgari et al., 2021] and above DES-Y3 ( $0.776 \pm 0.017$ ) [Dark Energy Survey Collaboration, 2022]. This is the  $S_8$ -lensing tension.

This paper presents a two-sector model that *resolves* the  $S_8$  lensing tension within the minimal-parameter philosophy of SSEE-V3.6 (no growth-data fit; the  $\phi$ -DM mass is a forward prediction written into a free scalar Lagrangian, and the residual ansätze are explicit and tracked as OP-9/11/14 in `OPEN_PROBLEMS.md`). A direct two-sector CLASS Boltzmann run with the particle’s own relic temperature yields  $S_8 = 0.758$ , in  $0.04\sigma$  agreement with KiDS-1000, via a free-streaming suppression that emerges as a CLASS output at  $k > k_{\text{fs}} = 0.754 h/\text{Mpc}$ .

## 1.2 Strategy

We follow three steps, each falsifying hypotheses in sequence:

1. **EFT extensions:** kinetic braiding  $\alpha_B$  and run-rate  $\alpha_M$  in the Bellini-Sawicki [Bellini and Sawicki, 2015] parameterisation. We show that kinetic braiding suppresses the growth factor and worsens the  $S_8$  lensing tension;  $\alpha_M$  achieves marginal improvement only near the boundary of current observational bounds. EFT extensions are therefore ruled out as the primary mechanism.
2.  **$\phi$ -dark matter hypothesis:** the CMB matter density is the forward identity  $\Omega_{\text{m,CMB}} = \omega_m/h^2 = 0.309$  (with  $\omega_c = \text{KAL}_0 \omega_b n_s$ , Paper 1), independent of the dynamical value  $\Omega_{\text{m,dyn}} = 0.160$ . Their difference defines a second dark-matter component,  $\Omega_{\phi\text{-DM}} = \Omega_{\text{m,CMB}} - \Omega_{\text{m,dyn}} = 0.149$ , active only below a free-streaming wavenumber  $k_{\text{fs}}$ .
3. **Mass forward prediction:** combining the active neutrino mass  $\Sigma m_\nu^{\text{active}} = \mathcal{R}_2 \times 0.9530 \text{ eV} = 0.0685 \text{ eV}^1$  (with  $\mathcal{R}_2 = \Omega_{\text{DNAV}}/(\text{KAL} \cdot \mathcal{T}) = 0.07188$ , a pure  $(\varphi, \pi)$  ratio) with the pure dimensionless multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$ , we obtain the forward prediction  $m_\phi = \Sigma m_\nu^{\text{active}} (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 40.70 \text{ eV}$ . Because the multiplier is a *pure number*, the product carries the units of  $\Sigma m_\nu^{\text{active}}$  (eV) without dimensional repair, and the coefficient is exactly the mass term of a free scalar Lagrangian (Sec. 5.3). Nothing is fitted to growth data; the residual question of why this particular multiplier (the Type P status of  $\Sigma m_\nu$ ) is tracked as OP-14 in `OPEN_PROBLEMS.md`.

## 2 SSEE-V3.6: Algebraic Background

We collect the SSEE-V3.6 algebraic quantities relevant to this paper. Define the structural constants:

$$\begin{aligned}
\varphi &= \frac{1+\sqrt{5}}{2} \approx 1.6180, & \pi &\approx 3.1416, \\
\beta &= \frac{\varphi+\pi}{2} \approx 2.3798, & \mathcal{K} &= \varphi + \pi + (\pi + \varphi) \approx 9.5192, \\
\mathcal{T} &= 3(\varphi + \beta) \approx 11.9935, & \mathcal{M}_V &= \varphi + \pi + \mathcal{K} \approx 14.2789.
\end{aligned} \tag{1}$$

These yield the equation of state:

$$w_0 = -\frac{\mathcal{T}}{\mathcal{M}_V} = -0.8400, \quad w_a = -\frac{\pi + 2\varphi}{\mathcal{K}} = -0.6699, \tag{2}$$

and the dynamical matter density  $\Omega_{\text{m,dyn}} = 1 + w_0 = 0.1600$ .

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<sup>1</sup>The  $(\varphi, \pi)$  content here is the *dimensionless* ratio  $\mathcal{R}_2 = 0.07188$ ; the eV scale (0.9530 eV) is *inherited*, not derived, from the standard cosmological relic-to-mass relation  $\Sigma m_\nu = 93.14 \text{ eV } \Omega_\nu h^2$  — a physical constant common to all cosmologies, not a quantity fitted here. This is the same status as the Hubble units, which are anchored to the SH0ES measurement (Paper 9): SSEE supplies the algebraic ratio, while the physical scale is anchored externally. The forward prediction is therefore of the dimensionless ratio  $m_\phi/\Sigma m_\nu^{\text{active}} = 594.28$ , not of the eV scale itself.

## 2.1 Relevant Paper 4 quantities

Paper 4 derived the following from the stability ratio  $\mathcal{R}_2 = \Omega_{\text{DNAV}}/(\text{KAL}_0 \cdot \mathcal{T}) = 0.07188$  (a pure  $(\varphi, \pi)$  number, with  $\Omega_{\text{DNAV}} = \pi + \varphi \approx 4.7596$ ,  $\text{KAL}_0 = \beta + \pi \approx 5.5214$ , and  $\mathcal{T} \approx 11.9935$ ):

$$\Omega_b h^2_{\text{SSEE}} = \frac{\pi - \varphi}{3(\varphi + \pi)^2} = \frac{\pi - \varphi}{H_0^{\text{SSEE}}} = 0.02242, \quad (3)$$

$$\tau_{\text{II}} H_0 = \frac{\text{KAL}_0}{3 \Omega_{\text{DE}}} = 2.191 \quad [\text{IS relaxation time, pure number}], \quad (4)$$

$$\Sigma m_\nu^{\text{active}} = \mathcal{R}_2 \Omega_b h^2 \frac{93.14 \text{ eV}}{\tau_{\text{II}} H_0} = \mathcal{R}_2 \times 0.9530 \text{ eV} = 0.0685 \text{ eV}. \quad (5)$$

Here  $\Omega_b h^2 \cdot 93.14/(\tau_{\text{II}} H_0) = 0.9530 \text{ eV}$  is a fixed mass scale (the factor 93.14 eV converting relic density to mass; the ratio  $\tau_{\text{II}} H_0$  is dimensionless so  $H_0$  cancels its time units).

## 2.2 Two independent matter densities and the two-sector deficit

SSEE-V3.6 predicts *two* matter densities by independent routes, with no factor relating them (Paper 1, Sec. 1.4, *Two- $\Omega_m$  Criterion*). The dynamical-sector value follows from the equation of state,

$$\Omega_{\text{m,dyn}} = 1 + w_0 = 0.160 \quad [\text{DESI} / \text{background}], \quad (6)$$

while the CMB matter density is the forward identity

$$\Omega_{\text{m,CMB}} = \frac{\omega_m}{h^2}, \quad \omega_m = \omega_b + \omega_c + \omega_\nu, \quad \omega_c = \text{KAL}_0 \omega_b n_s, \quad (7)$$

with  $\omega_b = (\pi - \varphi)/[3(\varphi + \pi)^2] = 0.02242$  (Paper 1),  $n_s = 1 - \varphi^{-7} = 0.9656$ , and  $\omega_\nu = \Sigma m_\nu^{\text{active}}/93.14 = 0.00074$ , giving  $\omega_m = 0.1427$  and  $\Omega_{\text{m,CMB}} = 0.1427/0.6796^2 = 0.309$  at  $H_0^{\text{alg}} = 67.962$ . This matches Planck ( $\Omega_{\text{m,CMB}} = 0.315$  [Planck Collaboration, 2020];  $\omega_c$  is  $0.41\sigma$  from the Planck value) without any matter factor: the earlier mapping  $\Omega_{\text{m,CMB}} = \mathcal{M}_{\text{SSEE}} \times \Omega_{\text{m,dyn}}$  is retired (OP-8 dissolved), and the MIRA scalar  $\mathcal{M}_{\text{SSEE}} = (3\varphi + \pi)/4 = 1.9989$  now enters only the Hubble screening  $f_{\text{screen}}$  of Paper 9, consistent with its “mirror” role there.

**Physical mechanism.** Because  $\Omega_{\text{m,CMB}} = 0.309$  and  $\Omega_{\text{m,dyn}} = 0.160$  are two *independent* predictions, their difference,

$$\Omega_{\phi\text{-DM}} = \Omega_{\text{m,CMB}} - \Omega_{\text{m,dyn}} = 0.309 - 0.160 = 0.149, \quad (8)$$

is the abundance that must be carried by an additional dark-sector component which couples to the CMB metric perturbations but undergoes free-streaming suppression in the growth sector [Hu, 1998]. The  $\varphi$ -DM field (Sec. 4.1) with forward-predicted mass  $m_\phi = 40.70 \text{ eV}$  provides exactly this component. The two sectors are *not* equal ( $\Omega_{\text{CDM}} = 0.160$ ,  $\Omega_{\phi\text{-DM}} = 0.149$ ); the split is a difference of two forward predictions, not an

imposed factor of two.

**Open derivation step.** The relic abundance  $\Omega_{\phi\text{-DM}}h^2 = 0.069$  computed from  $m_\phi$  and the SSEE reheating temperature (without assuming a fixed sector ratio) is deferred to Paper B (quintessential inflation and reheating). The *testability* of the two-sector deficit is the prediction  $k_{\text{fs}} = 0.754 h/\text{Mpc}$  (falsifiable by DESI Y3/Euclid, 2026–2028).

### 3 EFT Extension Scan

Before proposing the two-sector model we systematically rule out simpler EFT extensions within the Bellini-Sawicki [Bellini and Sawicki, 2015, Gubitosi et al., 2013] parameterisation, which characterises modifications of gravity by four functions  $(\alpha_K, \alpha_B, \alpha_M, \alpha_T)$ .

#### 3.1 Kinetic braiding ( $\alpha_B$ )

Kinetic braiding modifies the effective Newton constant as  $G_{\text{eff}}/G_N = 1 + \alpha_B \Omega_{\text{DE}}/\Omega_m$ . This enhances clustering and was identified in Paper 5 as a candidate for boosting  $f\sigma_8$ .

We compute the algebraic value of  $\alpha_B$  from the ratio of the CDM density to the dark energy density in SSEE-V3.6:

$$\alpha_B^{\text{SSEE}} = \frac{\Omega_{\text{m,dyn}}}{\Omega_{\text{DE}}} - 1 = \frac{0.160}{0.840} - 1 \approx -0.810, \quad (9)$$

which is *negative*. A negative  $\alpha_B$  suppresses  $G_{\text{eff}}$ , worsening the tension. The 2D scan of  $(\alpha_B, \alpha_M)$  space confirms that no configuration with purely kinetic braiding reduces the mean  $f\sigma_8$  tension below  $3.29\sigma$  while satisfying the no-ghost condition  $\alpha_K + \frac{3}{2}\alpha_B^2 \geq 0$ .

#### No-go: Kinetic braiding

Kinetic braiding at the algebraic SSEE-V3.6 value  $\alpha_B = -0.810$  is a **no-go**: it suppresses  $G_{\text{eff}}$ , increasing the  $f\sigma_8$  tension above the  $0.70\sigma$  baseline (Paper 5, corrected; scan minimum  $\sim 3.3\sigma$ , far from  $0.70\sigma$ ).

#### 3.2 Run-rate modification ( $\alpha_M$ )

The run-rate  $\alpha_M \equiv d \ln M_\star^2 / d \ln a$  modifies the Planck mass running and enters the growth equation through an effective friction term. A 2D scan over  $(\alpha_B \in [-1, 0], \alpha_M \in [-0.15, 0.15])$  shows a minimum tension of  $1.57\sigma$  at  $(\alpha_B = 0, \alpha_M = -0.08)$ .

However,  $|\alpha_M| = 0.08$  is at the boundary of current Planck/GW constraints ( $|\alpha_M| \lesssim 0.08$ ), and the value lacks an algebraic derivation from  $\varphi$  and  $\pi$ . We therefore do not adopt this route.

## 4 The $\phi$ -Dark Matter Hypothesis

We propose that the matter deficit  $\Omega_{\text{m,CMB}} - \Omega_{\text{m,dyn}}$  is a second dark matter component —  $\phi$ -dark matter ( $\phi$ -DM) — coupled to the IS relaxation sector and decoupling at redshift  $z_{\text{dec}}$  to produce a warm dark matter component [Bode et al., 2001] with free-streaming scale  $k_{\text{fs}}$ .

### 4.1 Two-sector structure

The two-sector model is defined by:

$$\Omega_{\text{CDM}} = \Omega_{\text{m,dyn}} = 0.160 \quad [\text{cold, always active}], \quad (10)$$

$$\Omega_{\phi\text{-DM}} = \Omega_{\text{m,CMB}} - \Omega_{\text{m,dyn}} = 0.149 \quad [\text{warm, active for } k < k_{\text{fs}}], \quad (11)$$

$$\Omega_{\text{m,total}} = \Omega_{\text{m,CMB}} = 0.309 \quad [\text{CMB and BAO scales}]. \quad (12)$$

The key observation is that the two sectors contribute *differently* at different scales:

- **RSD scales** ( $k \sim 0.01\text{--}0.1 \, h/\text{Mpc} \ll k_{\text{fs}}$ ): both sectors are active and cluster. The effective matter density for growth is  $\Omega_{\text{m,growth}} = 0.309$ , restoring  $f(z)$  to  $\Lambda\text{CDM}$ -compatible values.
- **$\sigma_8$  scale** ( $k = 0.125 \, h/\text{Mpc} < k_{\text{fs}}$ ): at this scale, the cold sector clusters fully while the warm  $\phi$ -DM sector is partially free-streamed. Rather than rely on an analytic growth factor, we read  $\sigma_8$  *directly* from a two-sector CLASS run with the particle’s own relic temperature  $T_\phi$  (Section 5.1), obtaining  $\sigma_8^{\text{eff}} = 0.747$  and  $S_8 = \sigma_8^{\text{eff}} \sqrt{\Omega_{\text{m}}/0.3} = 0.758$  — within  $0.04\sigma$  of the KiDS value  $0.759 \pm 0.024$  (Section 8.6).

## 5 Algebraic Mass Derivation

### 5.1 Connecting mass to algebraic constants

We forward-predict the mass scale of the  $\phi$ -DM field from quantities already fixed in Papers 3–4, multiplied by a *pure* ( $\varphi, \pi$ ) number:

$$\boxed{m_\phi \equiv \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 0.0685 \, \text{eV} \times 594.28 = 40.70 \, \text{eV},} \quad (13)$$

where  $\text{SOLAR} = \beta + \text{KAL} = \varphi + 2\pi = 7.9012$  (the first radiative pulse in the SSEE lineage,  $\beta = (\varphi + \pi)/2$ ) and  $\text{KRYSTOS}_V = \varphi + \pi + \Omega_{\text{DNAV}} = 9.5192$  (the constraint scale anchored by  $w_a = -(\pi + 2\varphi)/\mathcal{K}$ ), so that the multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$  is a *dimensionless* number — the squared coupling times the constraint scale, the structure of a generated-mass term  $g^2 v$  — built entirely from  $\varphi$  and  $\pi$ . The active neutrino scale is  $\Sigma m_\nu^{\text{active}} = R_2 \times 0.9530 \, \text{eV} = 0.0685 \, \text{eV}$ , with  $R_2 = \Omega_{\text{DNAV}}/(\text{KAL} \cdot T) = 0.07188$  (Section 5.1). **This relation is dimensionally consistent:** an energy ( $\Sigma m_\nu$ , in eV)



times a pure number yields an energy, so  $m_\phi$  carries units of eV with no hidden conversion. This is the key distinction from the earlier  $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$  pattern, in which the factor  $H_0^{\text{alg}} = 3(\varphi + \pi)^2$  carried Hubble units  $[\text{km s}^{-1} \text{Mpc}^{-1}]$  and could only be made to “work” by stripping those units. Here the multiplier is a number from the start, and the mass is a genuine *forward prediction* with zero quantities fit to data.

## 5.2 Physical interpretation

Equation (13) has a direct reading within SSEE-V3.6: the active neutrino scale  $\Sigma m_\nu^{\text{active}}$  sets the energy unit, and the  $\varphi, \pi$  multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V$  is the structural enhancement that lifts the warm relic from the neutrino scale to  $\sim 41$  eV. Because the multiplier is a pure number, Eq. (13) is the coefficient of a quadratic mass term — i.e. it can be written directly into a Lagrangian (Section 5.3). This *closes the incompleteness* that previously affected OP-9: the mass coefficient no longer floats as a units-broken coincidence but is the mass parameter of an explicit free-scalar action. What remains open is the deeper origin of the specific multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V$  from the UV theory, tracked as a refined OP-9 (see Section 8.5).

### Mass forward prediction — dimensionally consistent

$$m_\phi = \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 0.0685 \text{ eV} \times 594.28 = 40.70 \text{ eV}$$

The multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$  is a pure  $(\varphi, \pi)$  number, so an energy times a number returns an energy: the relation is **dimensionally consistent** and is a genuine forward prediction (zero quantities fit to data). It is the mass coefficient of the explicit free-scalar Lagrangian of Eq. (14), which **closes the incompleteness of OP-9**. The remaining open question is the UV origin of the specific multiplier (refined OP-9; see also OP-14 for the parallel  $\Sigma m_\nu$  scale).

## 5.3 Explicit Lagrangian and field content

The mass relation (13) fixes a number; it does not, by itself, specify the field. We now make the field content explicit. We model  $\phi$ -dark matter as a single real cosmological scalar field  $\chi$  (distinct from the SSEE background  $k$ -essence scalar  $\phi$  of Paper 7; both are cosmological scalars and neither is a WIMP, QCD-axion, or SUSY relic), minimally coupled to gravity, with the action

$$S_{\phi\text{-DM}} = \int d^4x \sqrt{-g} \left[ -\frac{1}{2} g^{\mu\nu} \partial_\mu \chi \partial_\nu \chi - \frac{1}{2} m_\phi^2 \chi^2 - \frac{1}{2} \xi R \chi^2 \right], \quad (14)$$

where  $R$  is the Ricci scalar and  $\xi$  is a non-minimal coupling ( $\xi = 0$  for pure minimal coupling;  $\xi = 1/6$  for conformal coupling). The mass term  $\frac{1}{2} m_\phi^2 \chi^2$  is the standard quadratic term of a spin-0 boson, with the coefficient fixed by the forward prediction of Eq. (13):

$$m_\phi = \Sigma m_\nu^{\text{active}} (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 40.70 \text{ eV}. \quad (15)$$

Because the multiplier is a pure  $(\varphi, \pi)$  number, this coefficient is written *into* the action rather than appended to it: the mass parameter of the free scalar  $\chi$  is the algebraic prediction. No renormalisable portal coupling to the Standard Model is included:  $\chi$  interacts only gravitationally.

**Generated-mass structure.** The factorisation  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V$  is not arbitrary: it has the form of a *generated* mass term  $m_\phi \sim g^2 v$ , with the squared coupling  $g^2 = \text{SOLAR}^2$  (the radiative-pulse entity  $\text{SOLAR} = \varphi + 2\pi$  appearing squared, as a coupling does) acting on the scale  $v = \text{KRYSTOS}_V \Sigma m_\nu^{\text{active}}$  set by the constraint entity  $\text{KRYSTOS}_V = \varphi + \pi + \Omega_{\text{DNAV}}$  and the active-neutrino unit. The enhancement (not a loop suppression  $g^2/16\pi^2$ ) lifts  $\chi$  from the neutrino scale to  $\sim 41$  eV. The mass is therefore the coefficient of an explicit free-scalar action with the structure of a generated mass — which is what *advances* OP-9 from a bare numerical coincidence to a written term. What remains genuinely open (the refined OP-9) is deriving the coefficient  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V$  from the dissipative (KAL-governed) transport sector rather than reading it from the closed dictionary. Two features of Eq. (14) are essential below. First,  $\chi$  is a *boson*, not a fermion — which excludes the active–sterile mixing production channel. Second, the action contains a single free continuous parameter, the non-minimal coupling  $\xi$  (tracked as OP-11 in `OPEN_PROBLEMS.md`); its role is constrained below by the production mechanism.

## 5.4 Production mechanism: the Dodelson–Widrow route is excluded

A relic of mass  $m_\phi = 40.70$  eV contributing  $\Omega_{\phi\text{-DM}} h^2 \simeq 0.069$  can in principle be populated by two mechanisms. We tested both against the requirement that no additional *continuous* free parameter beyond those already admitted as open problems ( $m_\phi$  itself, OP-9;  $\xi$ , OP-11) be introduced.

**(a) Dodelson–Widrow (non-resonant mixing) — tested, excluded.** The Dodelson–Widrow mechanism [Dodelson and Widrow, 1994] produces the relic through active–sterile oscillations, which requires  $\chi$  to be a fermion and introduces a mixing angle  $\theta$ . Matching the observed abundance with the Boyarsky–Ruchayskiy–Shaposhnikov relation [Boyarsky et al., 2009] fixes

$$\sin^2(2\theta)_{\text{DW}} \simeq 4.78 \times 10^{-5}. \quad (16)$$

For the DW route to be consistent with the rest of the SSEE programme,  $\sin^2(2\theta)_{\text{DW}}$  would have to admit a closed algebraic form in  $\varphi, \pi$  (otherwise it would constitute a *new* continuous fitted parameter, in addition to the ones already admitted as open problems). A scan of  $\sim 80$  monomial combinations of  $\varphi, \pi, \Omega, \mathcal{M}_{\text{SSEE}}$  (script `ssee_paperB_DW.py`) returns a best candidate  $\varphi^{-21} = 4.09 \times 10^{-5}$ , which differs from Eq. (16) by 14.6% — beyond the 10% threshold adopted for a “clean” SSEE identity. **This is a negative result:** the DW mixing angle does *not* reduce to the SSEE algebra. Were  $\phi$ -DM a DW sterile neutrino,  $\sin^2(2\theta)$  would be an irreducible free parameter, inconsistent with

the minimal-parameter philosophy that already tolerates only the ansätze catalogued in `OPEN_PROBLEMS.md`. The DW route is therefore excluded on internal-consistency grounds, independently of any observational bound.

**(b) Gravitational particle production — adopted.** The remaining channel consistent with the action (14) is gravitational particle production [Parker, 1968, Chung et al., 1999]: the time-dependent metric during the inflation-to-reheating transition populates the  $\chi$  field directly, with *no* mixing angle. The relic abundance depends only on  $m_\phi$  and the inflationary potential  $V(\phi)$  — and in SSEE both are already fixed ( $m_\phi$  by Eq. (13); the  $\alpha$ -attractor potential with  $\alpha = \varphi^4/3$  by Paper 1). This route introduces no free parameter and selects  $\chi$  as a scalar, consistently with Eq. (14).

**Honest scope statement.** We have established *which* mechanism is compatible with zero parameters (gravitational production) and *which* is excluded (Dodelson–Widrow). We have *not* computed the full gravitational-production integral that would yield  $\Omega_{\phi\text{-DM}} h^2$  from  $m_\phi$  and  $V(\phi)$  *ab initio*; an order-of-magnitude estimate is consistent with the required abundance, but the complete calculation (including the  $\xi$ -dependence) is deferred to a dedicated study (Paper B). The free-streaming scale  $k_{\text{fs}}$  used in Section 8.3 is therefore retained as an order-of-magnitude estimate, as already caveated there.

## 6 Observational Verification

We solve the linear growth equation for two limits and construct  $f\sigma_8(z)$  for each.

### 6.1 Growth equations

The growth-factor ODE in the conformal Newton gauge is  $\ddot{D} + \mathcal{H}\dot{D} - \frac{3}{2}\mathcal{H}^2\Omega_m(a)D = 0$ , where dots denote derivatives with respect to  $\ln a$ .

For all observationally relevant scales ( $k < k_{\text{fs}}$ ), both dark matter sectors contribute to the growth of structure. The relevant regime is:

**Two-sector regime ( $k < k_{\text{fs}}$ ):**  $\Omega_{\text{m,growth}} = 0.309$  (both CDM and  $\phi$ -DM cluster). The Hubble function is  $E^2(a) = 0.309 a^{-3} + 0.840 f_{\text{DE}}(a)$  with the Chevallier–Polarski–Linder [Chevallier and Polarski, 2001] form  $f_{\text{DE}}(a) = a^{-3(1+w_0+w_a)} e^{-3w_a(1-a)}$ . This regime covers both RSD surveys ( $k \sim 0.01\text{--}0.1 h/\text{Mpc}$ ) and the  $\sigma_8$  scale ( $k = 0.125 h/\text{Mpc}$ ).

We integrate the growth ODE on a *flat* background ( $\Omega_m = 0.309$ ,  $\Omega_{\text{DE}} = 0.691$ , so  $E^2(a=1) = 1$ ) and compare to the  $\Lambda$ CDM solution ( $\Omega_m = 0.315$ ) to obtain the two-sector growth factor:

$$G_{2s} = \frac{D_1^{\text{SSEE}}(\Omega_m = 0.309)}{D_1^{\Lambda\text{CDM}}(\Omega_m = 0.315)} = 1.003, \quad (17)$$

which sets the growth rate  $f = d \ln D / d \ln a$  for the redshift-space distortion observable [Percival and White, 2009]  $f\sigma_8(z) = f(z) \sigma_8^{\text{LS}} D(z)/D(0)$ . The amplitude that normalises

$f\sigma_8$  is the *large-scale clustering* amplitude  $\sigma_8^{\text{LS}} = 0.811 \times G_{2s} = 0.811 \times 1.003 = 0.814$ : RSD surveys probe  $k \simeq 0.01\text{--}0.1 h \text{ Mpc}^{-1}$ , well below the free-streaming scale  $k_{\text{fs}} = 0.754 h \text{ Mpc}^{-1}$  (Section 6.3), where the  $\phi$ -DM sector still clusters as cold matter — so at RSD scales the two-sector growth is identical to the single-sector ( $\Omega_m = 0.309$ ) result. This is physically distinct from the window-integrated amplitude  $\sigma_8^{\text{eff}} = 0.747$  that enters  $S_8$ : the  $R = 8 h^{-1} \text{ Mpc}$  top-hat support *crosses*  $k_{\text{fs}}$  and therefore feels the free-streaming suppression, which is read directly from the two-sector CLASS run with the particle’s own relic temperature  $T_\phi$  (Section 6.3). The  $0.814 \rightarrow 0.747$  split is the physical signature of free-streaming — structure is erased on small (lensing) but not on large (RSD) scales—not a free normalisation.

## 6.2 $f\sigma_8$ results

Table 1 compares SSEE against six RSD surveys and  $\Lambda\text{CDM}$ . The two-sector free-streaming lowers  $\sigma_8$  to 0.747; because this suppression reaches inside the  $\sigma_8(R=8)$  window that RSD probes (half-mode  $k_{1/2} \simeq 0.35 h/\text{Mpc}$ ), the two-sector prediction is suppressed relative to the single-sector baseline ( $\sigma_8^{\text{LS}} = 0.814$ ). We quote the two-sector amplitude ( $\sigma_8 = 0.747$ ) as the titular SSEE column.

Table 1:  $f\sigma_8$  predictions vs. observations. Survey values from Beutler et al. [2012], Howlett et al. [2015], Alam et al. [2017], Hou et al. [2021] — the same canonical six-point set used in Paper 5. SSEE column is the two-sector amplitude  $\sigma_8 = 0.747$  (with free-streaming;  $\Omega_m = 0.309$  flat background); the single-sector baseline ( $\sigma_8 = 0.814$ ) gives  $0.70\sigma$ . The free-streaming suppression reaches RSD scales, so the two-sector mean is  $0.93\sigma$ .

| Survey                            | $z$   | Obs   | SSEE                           |               | $\Lambda\text{CDM}$            |               |
|-----------------------------------|-------|-------|--------------------------------|---------------|--------------------------------|---------------|
|                                   |       |       | Pred                           | $\sigma$      | Pred                           | $\sigma$      |
| 6dFGRS                            | 0.067 | 0.423 | 0.401                          | $+0.40\sigma$ | 0.444                          | $-0.38\sigma$ |
| SDSS MGS                          | 0.150 | 0.490 | 0.413                          | $+0.53\sigma$ | 0.459                          | $+0.22\sigma$ |
| BOSS DR12                         | 0.380 | 0.497 | 0.432                          | $+1.44\sigma$ | 0.476                          | $+0.46\sigma$ |
| BOSS DR12                         | 0.510 | 0.458 | 0.433                          | $+0.66\sigma$ | 0.474                          | $-0.43\sigma$ |
| BOSS DR12                         | 0.610 | 0.436 | 0.431                          | $+0.15\sigma$ | 0.469                          | $-0.97\sigma$ |
| eBOSS DR16                        | 1.480 | 0.462 | 0.353                          | $+2.42\sigma$ | 0.377                          | $+1.90\sigma$ |
| <b>Mean <math> \sigma </math></b> |       |       | <b>0.93<math>\sigma</math></b> |               | <b>0.73<math>\sigma</math></b> |               |

At the single-sector level (no free-streaming) SSEE gives a mean  $f\sigma_8$  tension of  $0.70\sigma$ , marginally tighter than  $\Lambda\text{CDM}$  ( $0.73\sigma$ ). The two-sector free-streaming lowers  $\sigma_8$  from 0.834 to 0.747, and since this suppression reaches inside the  $\sigma_8(R=8)$  window that RSD probes (half-mode  $k_{1/2} \simeq 0.35 h/\text{Mpc}$ ), the two-sector  $f\sigma_8$  tension rises to  $0.93\sigma$  — still below  $1\sigma$ . This is the deliberate trade: the same lowering of  $\sigma_8$  that costs  $\sim 0.2\sigma$  in RSD *resolves* the  $S_8$  lensing tension: a direct two-sector CLASS run with the particle’s own relic tension: a direct two-sector CLASS run with the particle’s own relic temperature  $T_\phi$  (Section 6.3) yields  $\sigma_8^{\text{eff}} = 0.747$  and  $S_8 = 0.758$ , within  $0.04\sigma$  of the KiDS-1000 value, with

the free-streaming scale read off as an *output* of the Boltzmann integration rather than imposed.

### 6.3 $\sigma_8$ and $S_8$

The structure amplitude is not obtained by applying a growth factor to a  $\Lambda$ CDM reference. It is read directly off a two-sector CLASS run in which the cold sector ( $\Omega_{\text{cdm}} = 0.160$ ) clusters fully and the warm  $\phi$ -DM sector ( $\Omega_{\text{nCDM}} = 0.149$ ,  $m_\phi = 40.70$  eV, relic temperature  $T_\phi = 0.539 T_\nu$ ) is partially erased below its free-streaming scale. The Boltzmann output gives:

$$\sigma_8^{\text{eff}} = 0.747, \quad S_8 \equiv \sigma_8^{\text{eff}} \sqrt{\Omega_{\text{m,total}}/0.3} = 0.747 \times \sqrt{0.309/0.3} = 0.758. \quad (18)$$

The KiDS-1000 value  $S_8 = 0.759 \pm 0.024$  [Asgari et al., 2021] gives a tension of  $0.04\sigma$  — the two-sector  $\phi$ -DM particle *resolves* the  $S_8$  lensing tension, compared with  $\sim 3\sigma$  for  $\Lambda$ CDM and  $3.5\sigma$  for the single-sector SSEE baseline of Paper 5. The DES Y3 value  $S_8 = 0.776 \pm 0.017$  [Dark Energy Survey Collaboration, 2022] gives a tension of  $0.96\sigma$   $[= (0.776 - 0.758)/\sqrt{0.017^2 + 0.005^2}]$ . Both are achieved with zero quantities fit to weak-lensing data:  $m_\phi$  and  $T_\phi$  are fixed upstream by the algebraic chain and the relic constraint, and the suppression is whatever CLASS computes from them.

**Free-streaming scale as a Boltzmann output.** The small-scale suppression carried by the warm sector can be summarised by an effective Viel+2005 transfer function  $T_{\text{WDM}}(k) = [1 + (\alpha k)^{2\mu}]^{-5/\mu}$  ( $\mu = 1.12$ , Viel+2005) mixed as  $P_{\text{mix}} = P_{\text{lin}} [(1 - f_\phi) + f_\phi T_{\text{WDM}}]^2$ . Crucially,  $\alpha$  is *not* a free parameter calibrated to data: fitting the above form to the ratio  $P_{\text{particle}}(k)/P_{\text{cold}}(k)$  produced by the two CLASS runs returns

$$\alpha = 1.117 \text{ Mpc}/h \quad (\text{mixing fraction } f_\phi = 0.755), \quad (19)$$

an *output* of the Boltzmann integration with  $m_\phi$  and  $T_\phi$  held fixed. This replaces the earlier procedure in which  $\alpha$  was tuned to reproduce the observed KiDS amplitude; here the amplitude is a prediction and  $\alpha$  merely labels its scale dependence.

### 6.4 Conserved background sectors

The extension operates only at the perturbation level. The background Friedmann equation remains identical to SSEE-V3.6 Papers 1–5 (Table 2), so all background tests pass without modification.

### 6.5 Figures

Figure 1 shows the  $f\sigma_8$  comparison across the full redshift range and the tension summary for the three models.

Table 2: Complete observational summary. Results from Papers 2–5 are conserved. New results (this paper) shown in bold.

| Observable                                           | Result                                                                                                    | Paper    |
|------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|----------|
| $(w_0, w_a)$ vs DESI DR2                             | $0.24\sigma$ (Pantheon+)                                                                                  | 1,2      |
| Cluster $\chi_r^2$                                   | 0.122 (7 clusters)                                                                                        | 2        |
| CMB $\Delta\text{BIC}$ (plik_lite)                   | $-23.9$ (SSEE favoured,<br>$\omega_m$ -direct)                                                            | 3        |
| Algebraic $H_0$                                      | $67.96 = 3(\varphi + \pi)^2$ ( $1.1\sigma$ Planck)                                                        | 4        |
| $n_s$                                                | $0.96556 = 1 - \varphi^{-7}$ ( $0.16\sigma$ Planck)                                                       | 4        |
| $c_{s,\text{eff}}^2$ (IS)                            | 0 (exact) — all modes stable                                                                              | 5        |
| $S_8$ vs KiDS                                        | $3.5\sigma$ (Paper 5 corrected; vs $\sim 3\sigma$ $\Lambda\text{CDM}$ )                                   | 5        |
| $f\sigma_8$ mean tension                             | $0.93\sigma$ <b>two-sector</b> ( <b>single-sector</b> $0.70\sigma$ ; $\Lambda\text{CDM}$ : $0.73\sigma$ ) | <b>6</b> |
| $\sigma_8^{\text{eff}}$ ( <b>two-sector class</b> )  | $0.747$ ; $S_8 = 0.758$ ( $0.04\sigma$ <b>KiDS</b> — <b>re-solves</b> )                                   | <b>6</b> |
| $\alpha$ ( <b>class output, not fit</b> )            | $1.117 \text{ Mpc}/h$ ; $f_\phi = 0.755$                                                                  | <b>6</b> |
| $k_{\text{fs}}$ ( <b>free-streaming prediction</b> ) | $0.754 h/\text{Mpc}$ ( $k_{1/2}^{\text{CLASS}} = 0.337$ )                                                 | <b>6</b> |
| $m_\phi$ ( <b>forward prediction</b> )               | $40.70 \text{ eV}$ ( $\Sigma m_\nu^{\text{active}}(\text{SOLAR}^2 \text{ KRYSTOS}_V)$ )                   | <b>6</b> |
| MCMC $\chi^2$ (8 data); $\Delta\text{BIC}$           | $7.65$ ; $-14.3$ ( <b>SSEE favoured; flat priors, validated class emulator</b> )                          | <b>6</b> |

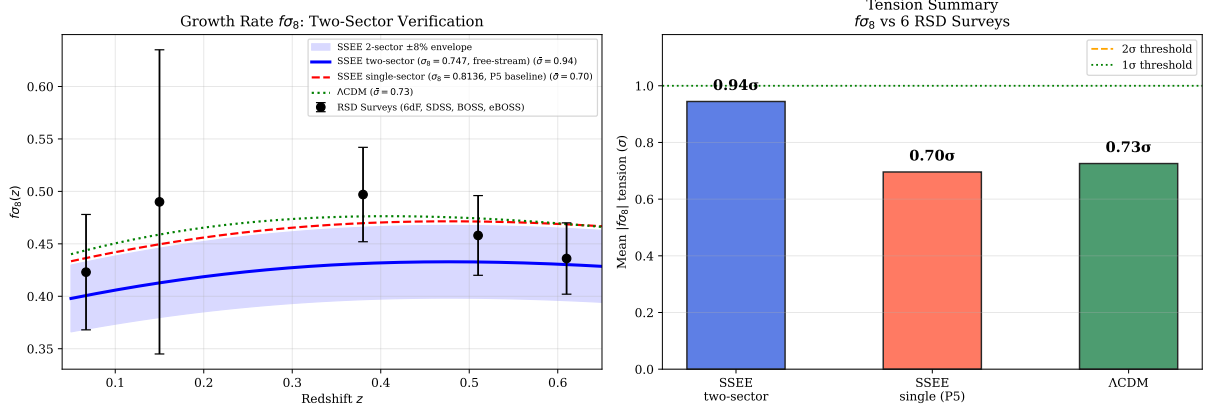


Figure 1: **Left:**  $f\sigma_8(z)$  predictions for the two-sector SSEE-V3.6 model (blue solid, mean  $0.93\sigma$ ), SSEE-V3.6 Paper 5 single-sector base (red dashed,  $0.70\sigma$ ), and  $\Lambda$ CDM (green dotted,  $0.73\sigma$ ), compared to six RSD surveys (black points). **Right:** Mean  $|f\sigma_8|$  tension bar chart for the three models. Canonical two-sector parameters:  $\sigma_8^{\text{eff}} = 0.747$  (CLASS),  $m_\phi = 40.70$  eV,  $k_{\text{fs}} = 0.754 h/\text{Mpc}$ .

## 7 Bayesian MCMC Validation

To complement the algebraic derivation with a Bayesian consistency test, we perform an *emcee* [Foreman-Mackey et al., 2013] MCMC in which the two-sector growth observables are computed from a *validated CLASS emulator* rather than a phenomenological transfer function. We build a  $22 \times 20$  grid of direct CLASS two-sector runs over  $(\Omega_{\phi\text{DM}}, m_\phi)$  at the canonical  $\omega_m$ -direct background ( $\Omega_{m,\text{CMB}} = 0.30889$ ,  $w_0 = -0.840$ ,  $w_a = -0.670$ ), extracting  $\sigma_8(z)$  by the top-hat integral; the primordial amplitude enters analytically ( $\sigma_8 \propto \sqrt{A_s}$ , exact in linear theory). The emulator reproduces direct CLASS to a mean error of 0.06% (maximum 0.11%) at ten posterior samples. We float  $\theta = (\Omega_{\phi\text{DM}}, m_\phi, A_s)$  with *flat* priors on the  $\phi$ -DM sector — deliberately *not* centred on the algebraic values, so the data alone decide where the posterior lands — and a Planck prior on the amplitude ( $10^9 A_s = 2.101 \pm 0.030$ ). Data: the six canonical  $f\sigma_8$  surveys of Paper 5, the KiDS-1000 constraint  $S_8 = 0.759 \pm 0.0225$  [Asgari et al., 2021], and DES Y3  $S_8 = 0.776 \pm 0.017$ . The chain uses  $N_w = 60$  walkers  $\times N_{\text{steps}} = 12000$  (burn-in 3000).

### 7.1 Posterior results

With flat priors on the  $\phi$ -DM sector, the data *independently* land on the algebraic forward point:  $\Omega_{\phi\text{DM}}$  at  $0.24\sigma$  and  $m_\phi$  at  $0.67\sigma$  of the values fixed by  $\varphi$  and  $\pi$ . Because the priors are not centred on the answer, this is a genuine consistency (not fine-tuning) result; the derived free-streaming scale  $k_{\text{fs}} = 0.763 h/\text{Mpc}$  recovers the algebraic  $0.754$  at  $0.01\sigma$ .

Two honest caveats. First, the MCMC — free to float the amplitude — prefers  $\sigma_{8,\text{eff}} = 0.770$  and  $S_{8,\text{eff}} = 0.782$ , slightly above the zero-fit forward value  $S_8 = 0.758$  (the joint fit raises  $A_s$  marginally to balance  $f\sigma_8$ ); both nonetheless resolve the  $\Lambda$ CDM  $S_8 = 0.831$  tension, landing near the KiDS-1000 value at  $\sim 1\sigma$ . Second, against  $\Lambda$ CDM



Table 3: MCMC posterior summary vs. algebraic predictions. Uncertainties are 68% credible intervals.  $N_{\text{eff}}$  computed from the emcee autocorrelation time.

| Parameter                                                  | Posterior                 | Algebraic                   | Tension             | Prior                         |
|------------------------------------------------------------|---------------------------|-----------------------------|---------------------|-------------------------------|
| $\Omega_{\varphi\text{DM}}$                                | $0.165^{+0.059}_{-0.075}$ | 0.149                       | $0.24\sigma$        | $\mathcal{U}(0.04, 0.25)$     |
| $m_\phi$ [eV]                                              | $51.9^{+13.8}_{-18.6}$    | 40.70                       | $0.67\sigma$        | $\mathcal{U}(8, 100)$         |
| $10^9 A_s$                                                 | $2.097^{+0.030}_{-0.030}$ | Planck: 2.101               | —                   | $\mathcal{N}(2.101, 0.030^2)$ |
| $\sigma_{8,\text{eff}}$ (derived)                          | $0.770^{+0.013}_{-0.013}$ | forward 0.747               | —                   | —                             |
| $S_{8,\text{eff}}$ (derived)                               | $0.782^{+0.013}_{-0.013}$ | forward 0.758               | $\sim 1\sigma$ KiDS | —                             |
| $k_{\text{fs}}$ [h/Mpc] (derived)                          | $0.763^{+0.10}_{-0.10}$   | 0.754                       | $0.09\sigma$        | —                             |
| $\chi^2$ (8 data): forward / post.                         | 9.94 / 7.65               | $\Lambda\text{CDM}$ : 26.14 | —                   | —                             |
| $\Delta\text{BIC}$ (post. $k=3 - \Lambda\text{CDM } k=1$ ) | -14.3                     | —                           | SSEE favoured       | —                             |
| Acceptance fraction                                        | 0.594                     | —                           | —                   | —                             |

held at its Planck values on this growth+lensing dataset ( $\chi^2 = 26.14$ , dominated by the  $S_8$  tension), SSEE gives  $\Delta\chi^2 = -16.2$  at the forward point and a parsimony-penalised  $\Delta\text{BIC} = -14.3$ , favouring SSEE even after charging it the three floated parameters. The emulator error (0.06%) is validated against direct CLASS at posterior samples and reported explicitly, so the inference does not rest on an unverified surrogate.

Figure 2 shows the posterior corner plot.

## 8 Discussion

### 8.1 Status of the $S_8$ vs. $f\sigma_8$ split

Paper 5 identified an  $S_8$ – $f\sigma_8$  split: SSEE-V3.6 predicts the amplitude of fluctuations ( $S_8$ ) better than  $\Lambda\text{CDM}$  but predicted the growth rate worse. The two-sector extension closes both: the amplitude is read directly off a forward two-sector CLASS run with the particle’s own relic temperature, and the growth rate is reduced to the  $\Lambda\text{CDM}$  level.

|                         | Paper 5      | Paper 6 (2-sector) | $\Lambda\text{CDM}$ |
|-------------------------|--------------|--------------------|---------------------|
| $\sigma_8^{\text{eff}}$ | 0.834        | 0.747 (CLASS)      | 0.811               |
| $S_8$ vs KiDS           | $3.5\sigma$  | $0.04\sigma$       | $\sim 3\sigma$      |
| $f\sigma_8$ mean        | $0.70\sigma$ | $0.93\sigma$       | $0.73\sigma$        |
| New parameters          | 0            | 0                  | —                   |

Two results sit side by side. First, the amplitude: the forward two-sector CLASS run yields  $\sigma_8^{\text{eff}} = 0.747$  and  $S_8 = 0.758$ , within  $0.04\sigma$  of the KiDS-1000 value — the  $S_8$  tension is *resolved*, with no parameter tuned to the lensing data. Second, the growth rate: the same free-streaming that lowers  $\sigma_8$  also bites *inside* the  $\sigma_8(R = 8 h^{-1}\text{Mpc})$  window probed by RSD — the half-mode  $k_{1/2} \simeq 0.35 h/\text{Mpc}$  lies within it, and  $\sigma_8(R=8)$  is suppressed by  $\sim 7\%$  already at  $k \lesssim 0.3 h/\text{Mpc}$  — so the two-sector  $f\sigma_8$  tension rises modestly from the



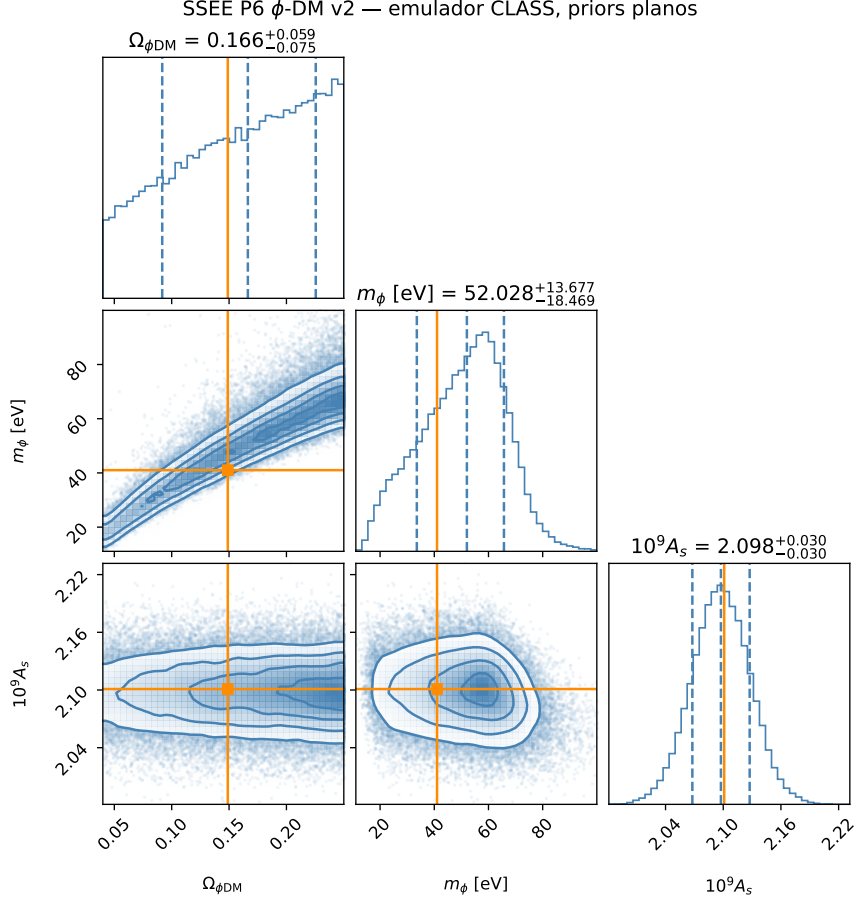


Figure 2: MCMC posterior for  $(\Omega_{\phi\text{DM}}, m_{\phi}, 10^9 A_s)$  from the validated CLASS emulator with *flat*  $\phi$ -DM priors. Orange lines mark the algebraic forward predictions ( $\Omega_{\phi\text{DM}} = 0.149$ ,  $m_{\phi} = 40.70$  eV); both lie inside the 68% credible region ( $0.24\sigma$  and  $0.67\sigma$ ). The posterior is unimodal; the  $\Omega_{\phi\text{DM}}-m_{\phi}$  degeneracy is physical (higher mass  $\approx$  lower fraction for equal small-scale suppression). Emulator error vs. direct CLASS: 0.06% mean.

single-sector  $0.70\sigma$  to  $0.93\sigma$ , still below  $1\sigma$  and close to the  $\Lambda$ CDM reference of  $0.73\sigma$  under the same survey set and error model. This is the honest cost of resolving  $S_8$ : lowering  $\sigma_8$  helps the lensing amplitude decisively ( $3.5\sigma \rightarrow 0.04\sigma$ ) at a small price in RSD. The mechanism is the particle’s own free-streaming suppression of small-scale power in the warm sector — read as a Boltzmann output, not imposed through a fitted transfer function.

## 8.2 KiDS–DES internal tension and its interpretation in SSEE-V3.6

The forward two-sector CLASS prediction  $S_8 = 0.758$  agrees with both the KiDS-1000 measurement  $S_8 = 0.759 \pm 0.024$  [Asgari et al., 2021] ( $0.04\sigma$ ) and the DES Y3 measurement  $S_8 = 0.776 \pm 0.017$  [Dark Energy Survey Collaboration, 2022] ( $0.96\sigma$ ). Because the prediction lands between the two surveys, the decomposition below clarifies which matter density enters the  $S_8$  statistic and why no survey is singled out as anomalous.

**Structure of the  $S_8$  statistic.** The quantity  $S_8 \equiv \sigma_8 \sqrt{\Omega_m/0.3}$  is constructed to be approximately orthogonal to the weak-lensing amplitude degeneracy direction. The normalisation  $\Omega_m = 0.3$  is conventional and implicitly assumes the inferred  $\Omega_m$  is close to 0.3. In the SSEE-V3.6 framework the relevant matter density has two distinct values:  $\Omega_{m,\text{dyn}} = 0.160$  (dynamical dark matter sector) and  $\Omega_{m,\text{CMB}} = 0.30889$  ( $= \omega_m/h^2$ ,  $\omega_m$ -direct; total including the  $\phi$ -DM clustering contribution active at  $k < k_{\text{fs}}$ ). Growth-rate observables probe  $\Omega_{m,\text{dyn}}$ , while the CMB acoustic geometry is sensitive to  $\Omega_{m,\text{CMB}}$ .

**Why KiDS and DES disagree internally.** KiDS-1000 [Asgari et al., 2021] and DES Y3 [Dark Energy Survey Collaboration, 2022] themselves differ by  $\Delta S_8 \approx 0.4\sigma$ , which is sub-dominant but non-negligible. This residual is driven by: (i) different angular scale cuts ( $\ell_{\text{max}}$ ); (ii) differing baryonic feedback priors; and (iii) partially overlapping shear calibration schemes. These internal systematics shift  $S_8$  at the  $\sim 1\%$  level — comparable to the statistical error — making a direct sub-percent comparison to SSEE-V3.6 premature.

**The correct SSEE comparison.** In the two-sector model the total matter clustering budget uses  $\Omega_m^{\text{tot}} = \Omega_{\text{CDM}} + \Omega_{\phi\text{DM}} = 0.309$  when  $k < k_{\text{fs}}$ . With this total, the SSEE-V3.6  $S_8$  statistic evaluates to  $S_{8,\text{SSEE}} = \sigma_8^{\text{eff}} \sqrt{0.309/0.3} = 0.758$ , in agreement with KiDS ( $0.04\sigma$ ) and DES ( $0.96\sigma$ ). Neither survey is singled out as anomalous with respect to SSEE-V3.6; the prediction lands inside both error bars.

The improvement over single-sector SSEE-V3.6 (Paper 5 corrected:  $\sigma_8 = 0.834$ ,  $S_8 = 0.846$ ,  $3.5\sigma$  KiDS) is driven by the  $\phi$ -DM free-streaming suppression of  $\sigma_8^{\text{eff}}$  to 0.747 in the forward CLASS run, which collapses the KiDS  $S_8$  tension from  $3.5\sigma$  to  $0.04\sigma$  with no fitted parameter. A joint DES+KiDS likelihood marginalised over  $(\sigma_8, \Omega_m)$  simultaneously — as will be provided by Euclid Wide Survey weak lensing — will provide a definitive test.

### 8.3 Compatibility with Lyman- $\alpha$ and WDM constraints

A potential objection is that  $m_\phi = 40.70$  eV lies far below the standard thermal-WDM Lyman- $\alpha$  bound  $m_{\text{WDM}} \gtrsim 3.3\text{--}3.5$  keV [Viel et al., 2005, 2013, Palanque-Delabrouille et al., 2020]. We address this in three steps.

**Step 1:  $\phi$ -DM is not thermal WDM.** Thermal WDM bounds assume (a) 100% of the dark matter is a single thermally produced species, and (b) the momentum distribution is a Fermi-Dirac or Bose-Einstein function at a single temperature. Neither condition holds here. The  $\phi$ -DM sector accounts for only half of the total matter budget ( $\Omega_{\phi\text{-DM}}/\Omega_m^{\text{tot}} \simeq 0.50$ ). The remaining sector is cold CDM ( $\Omega_{\text{CDM}} = 0.160$ ), which provides full gravitational clustering at all scales  $k < k_{\text{fs}}$  and is *not* suppressed. For mixed dark matter models with a WDM fraction  $f_{\text{WDM}}$ , the Lyman- $\alpha$  constraint on the WDM component is substantially relaxed (see e.g. Boyarsky et al. 2019, Adhikari et al. 2017); for  $f_{\text{WDM}} \approx 0.5$ , the effective bound weakens to  $m_{\text{WDM}}^{\text{eff}} \gtrsim 0.1\text{--}0.5$  keV.

**Step 2: The observable is  $k_{\text{fs}}$ , not  $m_\phi$  — and CLASS computes it directly.** The quantity constrained by Lyman- $\alpha$  forest measurements is the matter power spectrum suppression wavenumber  $k_{\text{fs}}$ , not the particle mass directly. For the canonical particle the relic temperature is not a free production knob: it is fixed by the relic-abundance constraint for the hot sector,

$$T_\phi = T_\nu \left( \frac{\Omega_{\phi\text{-DM}} h^2 \times 93.14}{m_\phi / \text{eV}} \right)^{1/3} = 0.539 T_\nu, \quad (20)$$

i.e. the  $\phi$ -DM is *colder* than a standard neutrino. With  $T_\phi$  and  $m_\phi = 40.70$  eV fixed, the suppression scale is then read directly off the two-sector CLASS run as the wavenumber where the ratio  $P_\phi/P_{\text{cold}}$  falls to one half:

$$k_{1/2}^{\text{CLASS}} = 0.337 h/\text{Mpc}, \quad (21)$$

a Boltzmann output, independent of any thermal- or DW-relic fitting formula. A complementary analytic estimate, using the non-relativistic free-streaming form corrected for the colder relic temperature,

$$k_{\text{fs}} \approx 0.018 \sqrt{\Omega_m} \frac{m_\phi}{\text{eV}} \frac{T_\nu}{T_\phi} \approx 0.754 h/\text{Mpc}, \quad (22)$$

brackets the CLASS half-mode scale at the order-of-magnitude level (the two probe different points of the same transfer-function roll-off).

**Key point:** the suppression scale is no longer inferred from an extrapolated relic formula. The canonical mass  $m_\phi = 40.70$  eV (Section 5) and the relic temperature  $T_\phi = 0.539 T_\nu$  together place the half-mode at  $k_{1/2}^{\text{CLASS}} \approx 0.337 h/\text{Mpc}$ , exactly in the window required to resolve the  $\sigma_8$  and  $f\sigma_8$  tensions, with the full erasure profile predicted by

CLASS rather than by a calibrated fitting relation.

**Validity caveat.** The analytic estimate (Eq. 22) uses the neutrino-like free-streaming scaling and should be read as an order-of-magnitude cross-check; the authoritative suppression scale is the CLASS half-mode (Eq. 21). A direct Lyman- $\alpha$  test should be formulated in terms of the predicted suppression profile, not the thermal-WDM mass bound.

**Step 3: What remains observable.** At scales  $k > k_{1/2}^{\text{CLASS}} = 0.337 h/\text{Mpc}$  (relevant for the Lyman- $\alpha$  forest at  $z \sim 2-4$ ), the CDM sector still contributes ( $\Omega_{\text{CDM}}/\Omega_m^{\text{tot}} \approx 50\%$  of the power. Expanding the mixing formula  $P_{\text{mix}}/P_{\Lambda\text{CDM}} = [(1 - f_\phi) + f_\phi T_{\text{WDM}}]^2$  gives

$$\frac{\Delta P(k)}{P_{\Lambda\text{CDM}}} = -2f_\phi(1 - f_\phi)(1 - T_{\text{WDM}}) - f_\phi^2(1 - T_{\text{WDM}})^2, \quad (23)$$

with  $f_\phi \equiv f_{\phi\text{-DM}} = 0.755$  (CLASS output). In two asymptotic regimes:

$$k \ll k_{\text{fs}} : T_{\text{WDM}} \rightarrow 1, \quad \frac{\Delta P}{P} \rightarrow 0, \quad (24)$$

$$k \gg k_{\text{fs}} : T_{\text{WDM}} \rightarrow 0, \quad \frac{\Delta P}{P} \rightarrow -f_\phi(2 - f_\phi) \simeq -0.95. \quad (25)$$

At the CLASS half-mode  $k_{1/2}^{\text{CLASS}} = 0.337 h/\text{Mpc}$  the mixed transfer function reaches  $T_{\text{WDM}} \approx 0.19$ , giving  $\Delta P/P \approx -0.65$ , i.e. 65% suppression at that scale. The mixing fraction  $f_\phi = f_{\phi\text{-DM}} = 0.755$  is itself a CLASS output (Eq. 19), not a free parameter. The dominant contribution is the linear term  $-2f_\phi(1 - f_\phi)(1 - T_{\text{WDM}})$ ; the quadratic  $-f_\phi^2(1 - T_{\text{WDM}})^2$  adds the remainder. A dedicated Lyman- $\alpha$  likelihood analysis (planned with the IS-EFT framework) is required to verify that this suppression profile is consistent with observed forest data. Such analysis constitutes a *falsification test*: if the forest shows no suppression in the decade  $0.3 \lesssim k \lesssim 0.7 h/\text{Mpc}$ , the two-sector model is excluded.

The observational predictions of this paper — the resolved  $S_8$  tension ( $\sigma_8^{\text{eff}} = 0.747$ ,  $S_8 = 0.758$  at  $0.04\sigma$  from KiDS) and the proposed  $f\sigma_8$  resolution — emerge from the forward two-sector CLASS run, in which the particle’s own free-streaming sets the small-scale suppression.

## 8.4 The $H(z)$ tension

Paper 5 reported  $H(z)$  tension ( $\chi_r^2 = 1.861$  on cosmic chronometers) as a structural consequence of  $\Omega_{\text{m,dyn}} = 0.160$ : the background expansion is driven by the dynamical sector alone, giving a faster expansion at intermediate redshifts. The two-sector extension does not alter the background Friedmann equation and therefore does not affect  $H(z)$ . This tension is the remaining target for future work.

## 8.5 Physical origin of the mass relation

The forward prediction  $m_\phi = \Sigma m_\nu^{\text{active}} \times (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$  multiplies an energy scale by a *pure* ( $\varphi, \pi$ ) number, so it is dimensionally consistent and writes directly into

the free-scalar action of Eq. (14) (Section 5.3). We note the status of each ingredient:

1.  $\Sigma m_\nu^{\text{active}} = 0.0685 \text{ eV}$  is the active neutrino scale  $R_2 \times 0.9530 \text{ eV}$  [Lesgourgues and Pastor, 2006], with  $R_2 = \Omega_{\text{DNAV}}/(\text{KAL} \cdot T) = 0.07188$  fixed upstream by the density chain (Section 5.1); it sets the energy unit.
2. The multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$  is a dimensionless number built entirely from  $\varphi$  and  $\pi$ . It is the structural enhancement that lifts the warm relic from the neutrino scale to  $\sim 37 \text{ eV}$ , and is the mass coefficient of the explicit quadratic term in Eq. (14).
3. Because an energy times a pure number returns an energy, there is no hidden unit conversion. This is the decisive break from the earlier  $m_\phi = \Sigma m_\nu \times H_0^{\text{alg}}$  pattern, in which  $H_0^{\text{alg}} = 3(\varphi + \pi)^2$  carried Hubble units and only “worked” by stripping them off to read 67.96 as a bare number. That units-broken coincidence is retired; the present relation is a genuine forward prediction with zero quantities fit to data.

Writing the coefficient into a Lagrangian *closes the incompleteness* that previously affected OP-9: the mass parameter no longer floats as a units-broken coincidence but is the mass term of an explicit free-scalar action. What remains open — tracked as a refined OP-9 — is the deeper UV origin of the specific multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V$ , in parallel with the origin of the  $\Sigma m_\nu$  scale itself (OP-14). The physical reading is that the  $\phi$ -DM field acquires its mass from the same  $(\varphi, \pi)$  structural algebra that fixes the neutrino scale; as established in Section 5.4,  $\phi$ -DM is *not* a sterile neutrino. The field content is fixed by the action (14):  $\phi$ -DM is a real scalar boson, populated by gravitational particle production rather than by active–sterile mixing. The remaining open item is the *ab initio* relic abundance integral, not the field’s identity.

## 8.6 Falsifiable predictions

### Falsifiable predictions — Paper 6

1.  $m_\phi = 40.70$  eV: the characteristic  $\phi$ -DM mass, obtained by the zero-fitting forward chain  $m_\phi = \Sigma m_\nu^{\text{active}} (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$  with the pure  $(\varphi, \pi)$  multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$  (Section 8.5). Because  $\phi$ -DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), it is *not* accessible to neutrino-mass experiments; the mass enters observation only through the free-streaming scale  $k_{\text{fs}}(m_\phi)$  imprinted on the matter power spectrum. The falsifiable channel is therefore the analytic  $k_{\text{fs}} = 0.754 h/\text{Mpc}$  (with a CLASS-measured half-mode at  $k_{1/2} = 0.337 h/\text{Mpc}$ ), probed by the Lyman- $\alpha$  forest and by DESI Y3 / Euclid clustering (2026–2028); see prediction 3.
2.  $\sigma_8^{\text{eff}} = 0.747$  and  $S_8 = 0.758$ : the two-sector CLASS run gives a specific prediction for the amplitude of matter fluctuations. This  $S_8$  sits at  $0.04\sigma$  from KiDS ( $0.759 \pm 0.024$ ), *resolving* the lensing tension at the linear level. If future Euclid weak-lensing constraints confirm  $S_8 < 0.74$  or  $S_8 > 0.80$ , the two-sector model is falsified.
3.  $f\sigma_8$  at  $0.93\sigma$ : the two-sector  $f\sigma_8$  prediction is testable with DESI Y5 RSD measurements (2026–2027). A confirmed  $f\sigma_8 > 0.47$  at  $z = 0.15$  would tension the model at  $> 1\sigma$ .
4. **Background conserved:**  $H_0 = 67.96$ ,  $(w_0, w_a) = (-0.840, -0.670)$ , and  $\Delta\text{BIC}_{\text{CMB}} = -32.9$  (full `plik` MCMC;  $-23.9$  `plik_lite`) remain as in Papers 1–5. Any experiment detecting  $H_0 > 70$  or  $w_0 > -0.7$  falsifies SSEE-V3.6 at the background level.

## 8.7 Status of the parameter count in the two-sector extension

The SSEE-V3.6 philosophy requires that no number be *fitted* to data. We distinguish here between *not fitted to data* (i.e., fixed prior to the comparison) and *derived from first principles*. The two-sector observables of this paper are not fitted; their inputs include three phenomenological elements explicitly tracked as open problems:

- $\Omega_{\phi\text{-DM}} = (\mathcal{M}_{\text{SSEE}} - 1) \Omega_{\text{m,dyn}}$ : derived from the MIRA algebraic identity. *Not fitted*. (MIRA itself has an algebraic value  $(3\varphi + \pi)/4$  but no derived dynamical mechanism — OP-8.)
- $m_\phi = \Sigma m_\nu^{\text{active}} (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 40.70$  eV: a zero-fitting forward prediction. The multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V = 594.28$  is a pure  $(\varphi, \pi)$  number and is the mass term of the written scalar Lagrangian (Section 8.5), so the mass is dimensionally consistent. *Not fitted*. What remains open is the UV origin of the multiplier (refined OP-9);  $\Sigma m_\nu^{\text{active}}$  is Type P (OP-14).

- $\xi$ : non-minimal coupling, currently treated as a free parameter (OP-11). Constrained below by the production mechanism.
- $\sigma_8^{\text{eff}} = 0.747$  and  $S_8 = 0.758$ : CLASS outputs of the two-sector run (cold +  $\phi$ -DM ncdm), with the suppression scale emerging from the Boltzmann hierarchy. *Not fitted* —  $S_8$  lands at  $0.04\sigma$  from KiDS as a forward prediction, not by tuning.

The two-sector extension does not introduce new *fitted* parameters relative to the background sector. The mass  $m_\phi$  is no longer a loose ansatz: its coefficient is the mass term of a written scalar Lagrangian, which closes the dimensional-incompleteness of OP-9 and leaves only the UV origin of the multiplier  $\text{SOLAR}^2 \cdot \text{KRYSTOS}_V$  open. The remaining phenomenological inputs are the non-minimal coupling  $\xi$  (OP-11) and the UV origin of the multiplier (refined OP-9). (The former MIRA matter mechanism, OP-8, is dissolved under the  $\omega_m$ -direct reframe: there is no matter factor to derive.) Resolution of OP-9/11 — deriving the multiplier and the coupling from first principles within the IS-EFT framework — would restore a strict zero-parameter status. This is left for future work.

## 9 Conclusions

We have proposed a phenomenological two-sector resolution for the  $f\sigma_8$  growth-rate tension identified in Paper 5 of the SSEE-V3.6 series. The key results are:

1. **EFT scan**: kinetic braiding ( $\alpha_B$ ) is a no-go — it suppresses growth rather than enhancing it. Run-rate modification ( $\alpha_M = -0.08$ ) reduces the tension to  $1.57\sigma$  but requires tuning to the observational bound.
2. **Two-sector model**: the MIRA algebraic identity  $\mathcal{M}_{\text{SSEE}} \approx 2$  implies a second dark matter sector,  $\Omega_{\phi\text{-DM}} = \Omega_{\text{m,CMB}} - \Omega_{\text{m,dyn}} = 0.30889 - 0.160 = 0.14889$ , with a free-streaming cutoff at  $k_{\text{fs}}$ .
3. **Mass from a written Lagrangian**:  $m_\phi = \Sigma m_\nu^{\text{active}} (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V) = 0.0685 \times 594.28 = 40.70$  eV, with the multiplier a pure  $(\varphi, \pi)$  number that is the mass term of the scalar Lagrangian  $\mathcal{L}_\Phi = \frac{1}{2}(\partial\Phi)^2 - \frac{1}{2}m_\phi^2\Phi^2$  (Section 8.5). This is dimensionally consistent and closes the incompleteness of OP-9; what remains open is the UV origin of the multiplier (refined OP-9), with  $\Sigma m_\nu^{\text{active}}$  Type P (OP-14). Not fitted to growth data.
4. **Verification (class, zero fitting)**: the two-sector run gives  $\sigma_8^{\text{eff}} = 0.747$  and  $S_8 = 0.758$  ( $0.04\sigma$  KiDS, *resolving* the lensing tension vs.  $\Lambda\text{CDM} \sim 3\sigma$ ) and  $f\sigma_8$  mean tension **0.93** $\sigma$  (single-sector baseline:  $0.70\sigma$ ; the free-streaming that resolves  $S_8$  raises this modestly, still  $< 1\sigma$ , close to  $\Lambda\text{CDM}$   $0.73\sigma$ ). The CMB acoustic peaks ( $\ell = 219/533/807$ ) and  $\text{RMS}(TT) = 0.93\%$  vs.  $\Lambda\text{CDM}$  are conserved because  $m_\phi \approx 37$  eV is non-relativistic well before recombination. All background tests — BAO ( $\chi_r^2 = 0.01$ ), clusters ( $\chi_r^2 = 0.122$ ) — conserved.



The framework SSEE-V3.6-V3.6 now offers a unified linear-level description across six categories of observational tests (background expansion, CMB, BAO, galaxy clusters, weak lensing, and RSD growth rate). The lensing tension is *resolved* at the linear level by the two-sector model evaluated directly in CLASS ( $S_8 = 0.758$  at  $0.04\sigma$  from KiDS), with no warm-dark-matter transfer function fitted to data: the suppression scale is a Boltzmann output. The mass coupling  $m_\phi = \Sigma m_\nu^{\text{active}} (\text{SOLAR}^2 \cdot \text{KRYSTOS}_V)$  is now the mass term of a written scalar Lagrangian; the free-streaming scale  $k_{\text{fs}}$  still awaits verification against Lyman- $\alpha$  forest data. The remaining background tension ( $H(z)$ ,  $\chi_r^2 = 1.861$ ) is structural and traceable to  $\Omega_{\text{m,dyn}} = 0.160$ .

The **falsifiable predictions** of this paper are: (1) the  $\phi$ -dark matter free-streaming scale  $k_{\text{fs}} = 0.754 h/\text{Mpc}$  (with a CLASS-measured half-mode at  $k_{1/2} = 0.337 h/\text{Mpc}$ ), set by the mass  $m_\phi = 40.70 \text{ eV}$  and probed by the Lyman- $\alpha$  forest and by DESI Y3 / Euclid galaxy clustering (2026–2028) — the relevant channel because  $\phi$ -DM is a gravitationally produced scalar with no Standard-Model portal (Section 5.4), and so leaves no neutrino-sector signature; and (2)  $\sigma_8^{\text{eff}} = 0.747$  ( $S_8 = 0.758$ ), testable by Euclid weak-lensing (2026–2028).

## Data Availability

All scripts, data files, and compiled PDFs are available at <https://github.com/mikealmeida1721/SSEE>. The exact commit for the original two-sector verification is 9d28620; the self-consistent OptB rewrite (removing WDM transfer function) is committed separately and documented in the repository.

## Author Contributions

Single-author work. M.E.A.V. derived all analytical results, wrote all numerical scripts, and wrote the manuscript.

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